

Supplementary Appendix

Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

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1 Model Structure and Assumptions

1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Clinical Definition	Utility	Monthly Cost
RF	Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses)	0.83	\$0
LR	Confirmed locoregional recurrence without distant metastasis; includes salvage surgery, re-treatment, and surveillance	0.64	\$3,000
DM	Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring	0.55	\$12,000
Death	All-cause mortality	0	\$0

Note: Utilities and costs are drawn from published melanoma CEA literature [1–3]. Costs are in 2026 USD. See Supplementary Table S9 for full PSA distribution parameters.

1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time t .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time t .
- **OS(t)** — Overall survival: probability of being alive at time t .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $RF = RFS$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $LR = DMFS - RFS$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $DM = OS - DMFS$.
- The dead state is the complement of overall survival: $Dead = 1 - OS$.

The constraint $RFS \leq DMFS \leq OS$ is enforced at all time points to ensure non-negative state occupancy. This ordering is clinically meaningful: a patient cannot have distant metastasis without having had a recurrence event ($RFS \leq DMFS$), and a patient cannot die without having both events ($DMFS \leq OS$). The ratio-constrained structure $r_{LR} = RFS/DMFS$ and $r_{DM} = DMFS/OS$ guarantees this ordering by construction.

1.3 Model Schematic

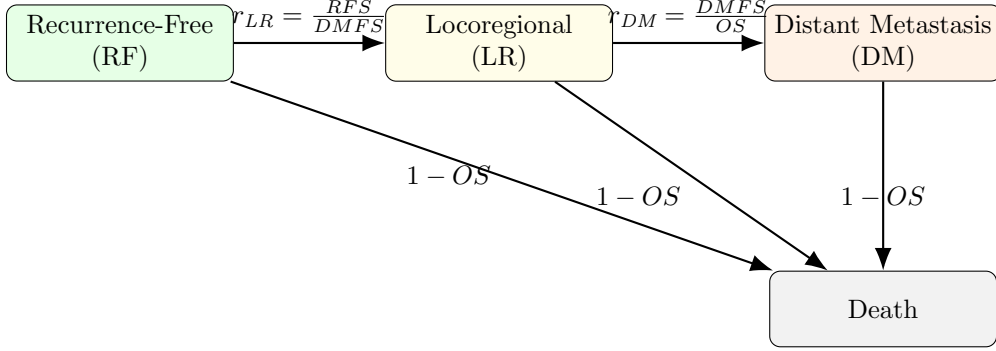


Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship $RFS \leq DMFS \leq OS$.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [4] using WebPlotDigitizer v4.7. Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS. The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual extraction. The 60-month landmark values match the published 5-year ASCO 2026 update [4]. At earlier time points, the 18-month RFS rates from the primary analysis [5] differ from digitized values by 1.4 pp (combo) and 0.2 pp (pembro), consistent with typical digitization error. The 5-year RFS 95% CIs: combo 56.3–78.3%, pembro 33.3–63.0%.

3 Clinical Input Provenance

Table S3 summarizes the provenance of all clinical inputs used in the model. The evidence status column classifies each input into one of five categories:

- **Published:** The value is directly reported in the cited source.
- **Digitized:** The value was extracted from a published Kaplan–Meier curve using Web-PlotDigitizer v4.7.
- **Derived:** The value is mathematically derived from other inputs (e.g., ratio of survival functions).
- **Modeled:** The value is produced by the model’s parametric extrapolation (e.g., long-term survival beyond observed data).
- **Topline only:** The source reports a statistically significant result but the numerical hazard ratio or landmark survival probabilities have not yet been disclosed.

Table S3: Clinical Input Provenance

Input	Value	Evidence Status	Source
Input	Value	Evidence Status	Source
<i>KEYNOTE-942 (Phase 2b, 5-year follow-up)</i>			
Recurrence-Free Survival (RFS)			
18-month RFS, combo	80.4%	Digitized	Fig. 1, Khattak 2026
24-month RFS, combo	78.3%	Digitized	Fig. 1, Khattak 2026
36-month RFS, combo	73.8%	Digitized	Fig. 1, Khattak 2026
48-month RFS, combo	72.4%	Digitized	Fig. 1, Khattak 2026
60-month RFS, combo	68.8%	Published	Khattak 2026
60-month RFS, combo 95% CI	56.3–78.3%	Published	Khattak 2026
18-month RFS, pembro	62.2%	Digitized	Fig. 1, Khattak 2026
24-month RFS, pembro	60.0%	Digitized	Fig. 1, Khattak 2026
36-month RFS, pembro	55.6%	Digitized	Fig. 1, Khattak 2026
48-month RFS, pembro	49.1%	Digitized	Fig. 1, Khattak 2026
60-month RFS, pembro	49.1%	Published	Khattak 2026
60-month RFS, pembro 95% CI	33.3–63.0%	Published	Khattak 2026
Distant Metastasis-Free Survival (DMFS)			
18-month DMFS, combo	92.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, combo	90.8%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, combo	85.6%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, combo	83.9%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, combo	–	Not reported	N/A
18-month DMFS, pembro	73.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, pembro	70.6%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, pembro	–	Not reported	N/A
Overall Survival (OS)			

Input	Value	Evidence Status	Source
18-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
24-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
36-month OS, combo	93.8%	Digitized	Fig. 1, Khattak 2026
48-month OS, combo	92.2%	Published	Khattak 2026
60-month OS, combo	92.2%	Published	Khattak 2026
18-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
24-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
36-month OS, pembro	85.6%	Digitized	Fig. 1, Khattak 2026
48-month OS, pembro	85.6%	Published	Khattak 2026
60-month OS, pembro	71.3%	Published	Khattak 2026
Hazard Ratios			
RFS HR (combo vs. pembro)	0.510	Published	Khattak 2026
RFS HR 95% CI	0.294–0.887	Published	Khattak 2026
DMFS HR (combo vs. pembro)	0.411	Published	Khattak 2026
DMFS HR 95% CI	0.200–0.843	Published	Khattak 2026
OS HR (combo vs. pembro)	0.471	Published	Khattak 2026
OS HR 95% CI	0.165–1.345	Published	Khattak 2026
OS events	14 (7 per arm)	Published	Khattak 2026
<i>INTerpath-001 (Phase 3, topline announcement)</i>			
RFS	Statistically significant	Topline only	Merck/Moderna 2026
DMFS	Statistically significant	Topline only	Merck/Moderna 2026
Numerical HR / KM curves	Not yet disclosed	Topline only	N/A
<i>Modeled / Derived Values</i>			
Fitted log-normal parameters			
OS μ (combo), σ (combo)	4.825, 0.879	Derived	Least-squares fit
OS μ (pembro), σ (pembro)	4.214, 1.332	Derived	Least-squares fit
r_DM μ (combo), σ (combo)	4.285, 0.851	Derived	Ratio fit
r_DM μ (pembro), σ (pembro)	3.475, 1.390	Derived	Ratio fit
r_LR μ (combo), σ (combo)	3.420, 1.250	Derived	Ratio fit
r_LR μ (pembro), σ (pembro)	2.268, 1.749	Derived	Ratio fit
Long-term extrapolation			
10-year OS, combo	74.4%	Modeled	Log-normal extrapolation
10-year OS, pembro	46.3%	Modeled	Log-normal extrapolation
20-year OS, combo (GP-constrained)	18.3%	Modeled	GP hazard floor
20-year OS, pembro (GP-constrained)	7.8%	Modeled	GP hazard floor
40-year OS, combo (GP-constrained)	0.0%	Modeled	GP hazard floor

3.1 Observed vs. Fitted Survival Curves

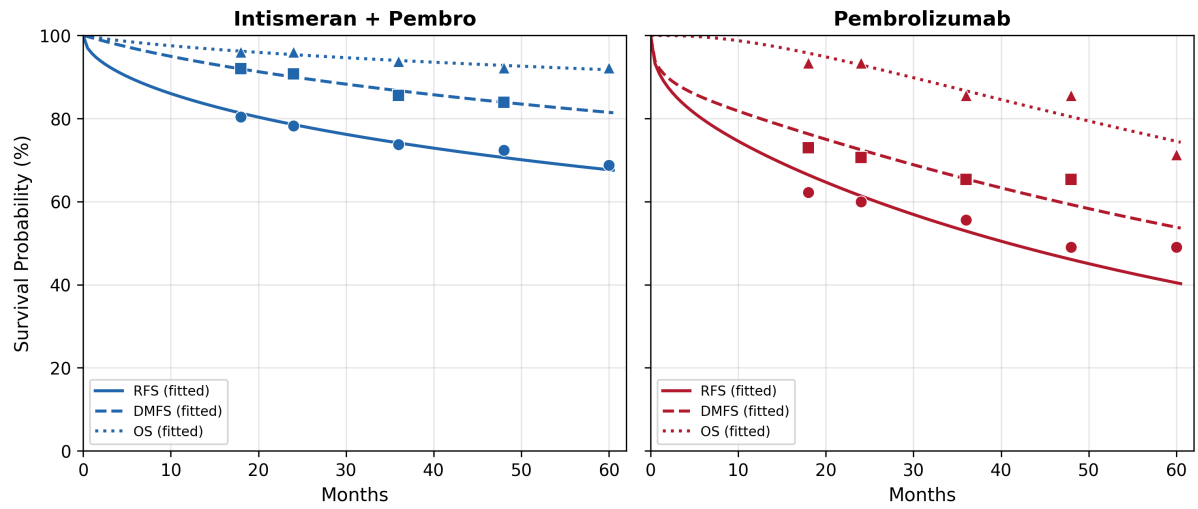


Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm.

4 Survival Model Selection

4.1 Component-Level Parametric Fit

Table S4: Component-Level Parametric Fit Across Five Distributions

Component	Distribution	AIC	RMSE	Rank
<i>OS – Combo arm</i>				
Exponential	1 param	-45.49	0.0087	4
Weibull	2 param	-49.84	0.0046	2
Log-normal	2 param	-50.08	0.0045	1
Log-logistic	2 param	-49.89	0.0046	3
Generalized gamma	3 param	-47.97	0.0045	5
<i>OS – Pembro arm</i>				
Exponential	1 param	-31.27	0.0359	4
Weibull	2 param	-32.00	0.0273	1
Log-normal	2 param	-31.16	0.0297	3
Log-logistic	2 param	-31.69	0.0282	2
Generalized gamma	3 param	-30.33	0.0264	5
<i>r_{DM} – Combo arm</i>				
Exponential	1 param	-36.79	0.0078	3
Weibull	2 param	-36.53	0.0063	4
Log-normal	2 param	-36.93	0.0060	1
Log-logistic	2 param	-36.61	0.0062	2
Generalized gamma	3 param	-34.74	0.0061	5
<i>r_{DM} – Pembro arm</i>				
Exponential	1 param	-19.41	0.0688	5
Weibull	2 param	-28.43	0.0174	3
Log-normal	2 param	-27.23	0.0202	4
Log-logistic	2 param	-30.30	0.0137	2
Generalized gamma	3 param	-32.45	0.0082	1
<i>r_{LR} – Combo arm</i>				
Exponential	1 param	-30.38	0.0392	5
Weibull	2 param	-38.97	0.0136	3
Log-normal	2 param	-39.49	0.0129	2
Log-logistic	2 param	-39.70	0.0126	1
Generalized gamma	3 param	-38.43	0.0118	4
<i>r_{LR} – Pembro arm</i>				
Exponential	1 param	-31.82	0.0340	5
Weibull	2 param	-33.60	0.0233	1
Log-normal	2 param	-33.37	0.0238	2
Log-logistic	2 param	-33.51	0.0235	3
Generalized gamma	3 param	-31.69	0.0231	4

4.2 Joint Model Selection

A total of 4,096 distribution combinations were evaluated for structural validity at 40 years. All 4,096 satisfied the $RFS \leq DMFS \leq OS$ ordering constraint (0 crossing violations). The full results are available online ([output/survival_combos_4096.csv](#)).

Table S5: Top 10 Joint Model Combinations by Total AIC

Rank	OS	r_{DM}	r_{LR}	Total AIC	OS40c	OS40p
	C/P	C/P	C/P			
1	LN / W	LN / LL	LL / W	-222.61	76.3%	0.0%
2	LN / W	LN / LL	LL / LL	-222.52	76.3%	0.0%
3	LN / W	Exp / LL	LL / W	-222.47	76.3%	0.0%
4	LLog / W	LN / LL	LL / W	-222.42	72.9%	0.0%
5	LN / W	LN / LL	LN / W	-222.40	76.3%	0.0%
6	LN / W	LN / LL	LL / LN	-222.39	76.3%	0.0%
7	LN / W	Exp / LL	LL / LL	-222.38	76.3%	0.0%
8	W / W	LN / LL	LL / W	-222.37	71.1%	0.0%
9	LLog / W	LN / LL	LL / LL	-222.33	72.9%	0.0%
10	LN / W	LN / LL	LN / LL	-222.31	76.3%	0.0%

Distributions: LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential. C/P = Combo/Pembro. All top-ranked combinations use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3% under unconstrained extrapolation, substantially higher than GP survival at age 101 ($\approx 0\%$). This motivated the GP mortality hazard floor (Section 5.1), under which the constrained OS40c falls to 0.0% for both arms.

4.3 Lifetime OS Extrapolation by Distribution

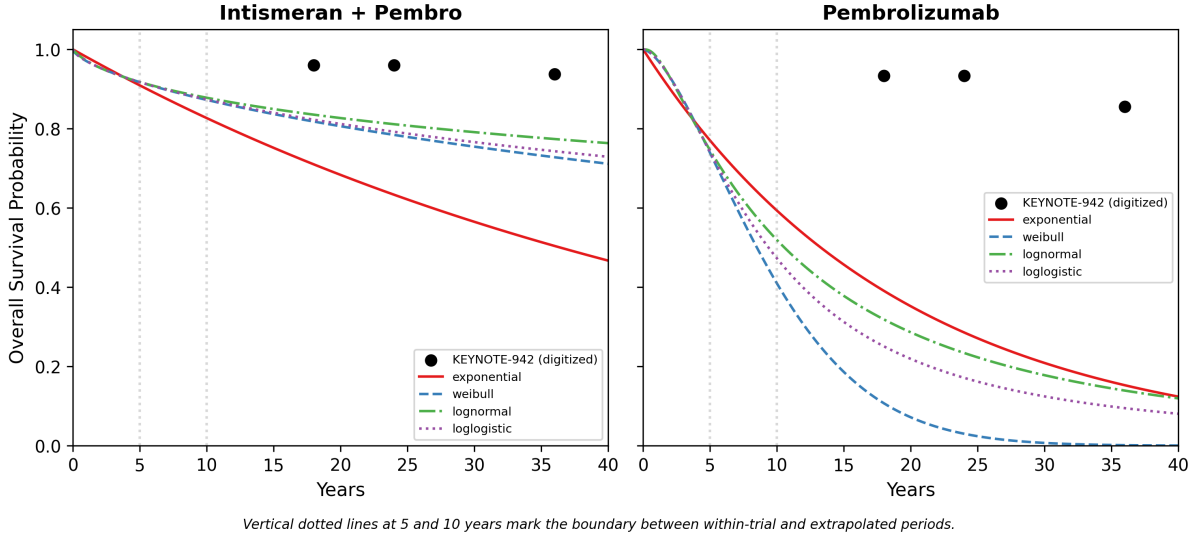


Figure S3: Lifetime OS extrapolation by parametric distribution. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; beyond 10 years is pure extrapolation. The four distributions produce wildly divergent 40-year survival estimates (0% to 76.3%).

5 General-Population Mortality and Structural Validation

5.1 GP Mortality Implementation

The GP mortality hazard floor is applied to modeled OS from month 60 onward.

Life table source: US 2019 period life tables (Arias et al., NVSS). **Baseline age:** 61 years (KEYNOTE-942 median). **Sex weighting:** 64% male, 36% female. **Interpolation:** Annual life table survival linearly interpolated to monthly resolution; monthly hazards derived as $h_{GP}(t) = -\ln[S_{GP}(t+1)/S_{GP}(t)]$.

Constraint formula (mortality hazard floor): from month 60 onward, the model mortality hazard is floored at the age- and sex-matched general-population hazard:

$$h_{\text{model}}^{\text{con}}(t) = \max[h_{\text{param}}(t), h_{\text{GP}}(t)] \quad t \geq 60 \text{ months}, \quad (1)$$

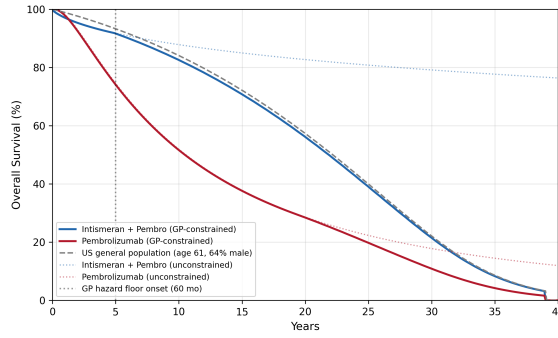
where $h_{\text{param}}(t) = -\ln[S_{\text{param}}(t+1)/S_{\text{param}}(t)]$ is the monthly hazard implied by the fitted parametric survival curve. The constrained survival is reconstructed as $S_{\text{model}}^{\text{con}}(t) = \exp[-\sum_0^t h^{\text{con}}(u)]$. This is a **hazard floor**, not a survival cap: a cohort that has already experienced elevated (cancer-related) mortality in the first 5 years remains below general-population survival at every subsequent time point, whereas a survival cap ($S = \min[S_{\text{param}}, S_{\text{GP}}]$) would incorrectly force the two curves to coincide whenever the parametric tail exceeded the general-population level. Both arms are subject to the same background mortality floor.

5.2 Long-Term Survival Diagnostics

Table S6: Long-Term Survival Predictions vs. General Population

Year	Combo OS	Pembro OS	GP Survival	Combo vs GP
5	91.6%	74.1%	93.3%	Below (excess mortality to date)
10	82.5%	51.6%	84.1%	Below (hazard floor active)
15	70.8%	37.6%	72.2%	Below (hazard floor active)
20	56.1%	28.4%	57.3%	Below (hazard floor active)
30	21.5%	10.9%	22.1%	Below (hazard floor active)
40	0.0%	0.0%	0.0%	At GP (age 101, GP survival ≈ 0)

5.3 OS versus General-Population Survival



GP-constrained curves (solid) use a mortality hazard floor from month 60. The constrained combo OS stays below the GP curve at all time points, reflecting the excess mortality experienced in the first 5 years. Unconstrained log-normal (dotted) produces implausibly high 40-year survival (76.4%).

Figure S4: OS versus general population. Solid lines: GP-constrained model (hazard floor from 60 months). Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US GP survival from age 61. The constrained combo curve remains below the GP curve at all time points, reflecting the hazard-floor property: a cohort with prior excess mortality does not "catch up" to general-population survival.

5.4 Structural Quality Assurance

Table S7: Structural Quality Assurance (480 Monthly Cycles)

Check	Result
Total cycles	480
RFS > DMFS violations	0
DMFS > OS violations	0
Negative RF state probability	0
Negative LR state probability	0
Negative DM state probability	0
Maximum state-sum error	$< 10^{-12}$
Clamp events (forced non-negative)	0

6 External Validation

6.1 External Validation Landmarks

Table S8: External Validation of Pembrolizumab Arm Predictions

Outcome	Year	Model (pembro)	KEYNOTE-054	CheckMate 238
RFS	5	40.5%	—	51% [6]
RFS	7	31.9%	50% [7]	46% [6]
DMFS	5	53.8%	—	59% [6]
DMFS	7	44.9%	54% [7]	55% [6]
OS	5	74.5%	NR	76% [8]
OS	7	64.2%	NR	71% [6]

NR: Not reported. CheckMate 238 evaluated nivolumab (not pembrolizumab). The model predicts lower RFS/DMFS versus external trials, reflecting the higher-risk KEYNOTE-942 population (stage IIIB–IV including resected stage IV melanoma vs IIIA–C in KEYNOTE-054 and IIIB–IV in CheckMate 238). Only published landmark values are included; no extrapolated estimates are used.[6, 7]

6.2 Longitudinal External Validation

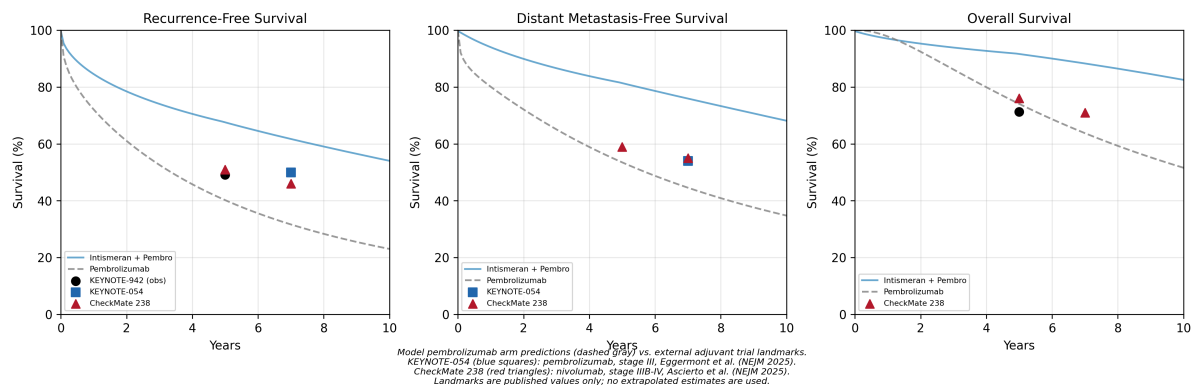


Figure S5: External validation of pembrolizumab arm predictions. Gray line: model. Circles: KEYNOTE-942 observed data. Squares: KEYNOTE-054 (pembrolizumab, 7-year). Triangles: CheckMate 238 (nivolumab, 5- and 7-year). Only published landmark values are shown.

7 Model Inputs

7.1 Complete Parameter Table

Table S9: Full Model Parameter Inventory

Parameter	Base	DSA Low	DSA High	PSA Distribution	Source
<i>Clinical Parameters</i>					
OS μ – combo	8.268	4.228	4.628	Normal (SE from Jacobian)	Fitted
OS σ – combo	0.478	0.378	0.578	Normal (SE from Jacobian)	Fitted
OS μ – pembro	3.725	3.525	3.925	Normal (SE from Jacobian)	Fitted
OS σ – pembro	0.574	0.474	0.674	Normal (SE from Jacobian)	Fitted
r _{DM} μ – combo	4.107	3.907	4.307	Normal (SE from Jacobian)	Fitted
r _{DM} σ – combo	0.432	0.332	0.532	Normal (SE from Jacobian)	Fitted
r _{DM} μ – pembro	2.849	2.649	3.049	Normal (SE from Jacobian)	Fitted
r _{DM} σ – pembro	0.934	0.834	1.034	Normal (SE from Jacobian)	Fitted
r _{LR} μ – combo	3.367	3.167	3.567	Normal (SE from Jacobian)	Fitted
r _{LR} σ – combo	0.624	0.524	0.724	Normal (SE from Jacobian)	Fitted
r _{LR} μ – pembro	2.750	2.550	2.950	Normal (SE from Jacobian)	Fitted
r _{LR} σ – pembro	0.638	0.538	0.738	Normal (SE from Jacobian)	Fitted
Treatment duration (pembro)	18 cycles (2 yr)	9	24	Fixed	[4]
Treatment duration (intismeran)	9 doses (1 yr)	6	12	Fixed	[4]
Baseline age	61 years	55	67	Fixed	[4]
Male proportion	64%	–	–	Fixed	[4]
<i>Costs (2026 USD)</i>					
Intismeran (one-time)	\$200,000	\$160,000	\$240,000	Gamma (CV=0.2)	Assumed
Pembrolizumab (annual)	\$220,896	\$176,717	\$265,075	Gamma (CV=0.2)	WAC (CMS)
Tumor sequencing	\$1,000	\$500	\$2,000	Gamma (CV=0.2)	CMS
Administration (per cycle)	\$500	\$250	\$750	Gamma (CV=0.2)	CMS
AE management (one-time)	\$15,000	\$7,500	\$22,500	Gamma (CV=0.2)	[3]
LR management (monthly)	\$3,000	\$2,400	\$3,600	Gamma (CV=0.2)	[3]
DM management (monthly)	\$12,000	\$9,600	\$14,400	Gamma (CV=0.2)	[1]
<i>Utilities</i>					
RF state	0.83	0.80	0.86	Beta (SE=0.03)	[2]
LR state	0.64	0.61	0.67	Beta (SE=0.03)	[2]
DM state	0.55	0.52	0.58	Beta (SE=0.03)	[2]
AE disutility (one-time)	0.05	0.00	0.10	Beta (SE=0.02)	Assumed
<i>Economic Parameters</i>					
Discount rate (costs, effects)	3%	0%	5%	Fixed (DSA only)	[9]
Time horizon	40 years	10	20	Fixed (scenario)	Assumed
Willingness-to-pay threshold	\$100K–\$150K/QALY	\$50K	\$200K	Fixed (scenario)	[10]
Cost base year	2026 USD	–	–	–	Assumed
GP mortality source	US 2019 Life Tables	–	–	–	NVSS (Arias 2022)

8 Additional Results

8.1 Lifetime Cost Decomposition

Table S10: Lifetime Cost Decomposition by Component

Component	Combo	Pembro	Incremental
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$493	\$493	\$0
Adverse events	\$15,000	\$0	\$15,000
LR management	\$112,203	\$74,477	\$37,726
DM management	\$524,121	\$453,082	-\$71,039
Total	\$1,070,705	\$745,939	\$324,765

The upfront \$200,000 intismeran cost is partially offset by \$71,039 in lower DM management costs, reflecting the DMFS HR of 0.411.

8.2 LY and QALY Decomposition

Table S11: Life-Year and QALY Decomposition by Health State

Outcome	Combo	Pembro	Incremental
Life-years (undiscounted)	23.78	15.34	7.13
Life-years (discounted 3%)	19.01	10.75	8.26
RF state QALYs	10.17	4.60	5.57
LR state QALYs	1.99	1.32	0.67
DM state QALYs	2.00	1.73	-0.32
Total QALYs	14.16	7.65	6.52

Health benefit is driven by RF state QALYs (5.57 gained), more than compensating for fewer DM-state QALYs (-0.32).

9 Uncertainty and Structural Scenario Analyses

9.1 Deterministic Sensitivity Analysis

Table S12: One-Way Deterministic Sensitivity Analysis

Parameter	Base	Low	High	ICER range	Change in NMB
Utility RF	0.83	0.80	0.86	51,431–48,346	\$60,441
Utility LR	0.64	0.61	0.67	50,082–49,601	\$9,431
Utility DM	0.55	0.52	0.58	49,954–49,728	\$4,440
Pembrolizumab annual cost	\$220,896	\$176,717	\$265,075	49,841–49,841	\$0
Intismeran cost	\$200,000	\$160,000	\$240,000	43,702–55,979	\$80,000
LR monthly cost	\$3,000	\$2,400	\$3,600	48,683–50,999	\$15,090
DM monthly cost	\$12,000	\$9,600	\$14,400	47,660–52,021	\$28,416
Discount rate	3%	0%	5%	49,330–53,363	\$798,812
OS μ combo	8.396	5.037	11.754	51,465–49,690	\$627,624
OS μ pembro	4.841	3.873	5.810	60,695–Dominant	\$494,048

The ICER remained below \$61,000/QALY across all tested ranges, with the combination dominant (lower cost and higher QALY) when the pembrolizumab OS parameter was increased by 20%.

9.2 Probabilistic Sensitivity Analysis

9.2.1 PSA Implementation Details

PSA with 6,000 Monte Carlo iterations (seed = 42). Survival parameters (μ , σ for 6 curves) drawn jointly from bivariate normal via Jacobian covariance ($r > 0.95$). Costs: gamma (CV=0.2). Utilities: beta (SE=0.03). Structural constraints (RFS $\leq$ DMFS $\leq$ OS, monotonicity, survival within [0,1]) verified every cycle; all 6,000 draws were structurally valid (0 rejections). No iterations were excluded on the basis of the ICER value.

Table S13: PSA Parameter Distributions and Sampled Values

Parameter	Distribution	Fitted mean	SE	Correlation	Sampling
OS μ / σ – combo	Bivariate normal	8.396 / 3.096	0.793 / 0.522	0.996	Joint
OS μ / σ – pembro	Bivariate normal	4.841 / 1.131	0.291 / 0.310	0.955	Joint
r _{DM} μ / σ – combo	Bivariate normal	7.113 / 2.480	0.814 / 0.563	0.995	Joint
r _{DM} μ / σ – pembro	Bivariate normal	7.042 / 5.000	2.278 / 3.130	0.995	Joint
r _{LR} μ / σ – combo	Bivariate normal	8.874 / 5.000	2.039 / 1.928	0.997	Joint
r _{LR} μ / σ – pembro	Bivariate normal	5.913 / 2.674	0.757 / 0.879	0.984	Joint
Intismeran cost	Gamma (CV=0.2)	\$200,000	—	—	Independent
Pembrolizumab annual cost	Gamma (CV=0.2)	\$220,896	—	—	Independent
LR monthly cost	Gamma (CV=0.2)	\$3,000	—	—	Independent
DM monthly cost	Gamma (CV=0.2)	\$12,000	—	—	Independent
Utility RF / LR / DM	Beta (SE=0.03)	0.83 / 0.64 / 0.55	—	—	Independent

9.2.2 PSA Results

- Mean Δ QALY: 6.65 (95% CI 4.13–9.83).
- Mean Δ Cost: \$293,238 (95% CI –\$533,282 to \$825,708).
- Mean NMB: \$371,570 at \$100K and \$703,975 at \$150K.
- Probability cost-effective: 52.4% at \$50K, 93.4% at \$100K, and 98.4% at \$150K.
- No iterations were excluded based on the ICER; 11.2% of draws were cost-saving, and all 6,000 draws had Δ QALY $\geq$ 0.

9.2.3 PSA Convergence

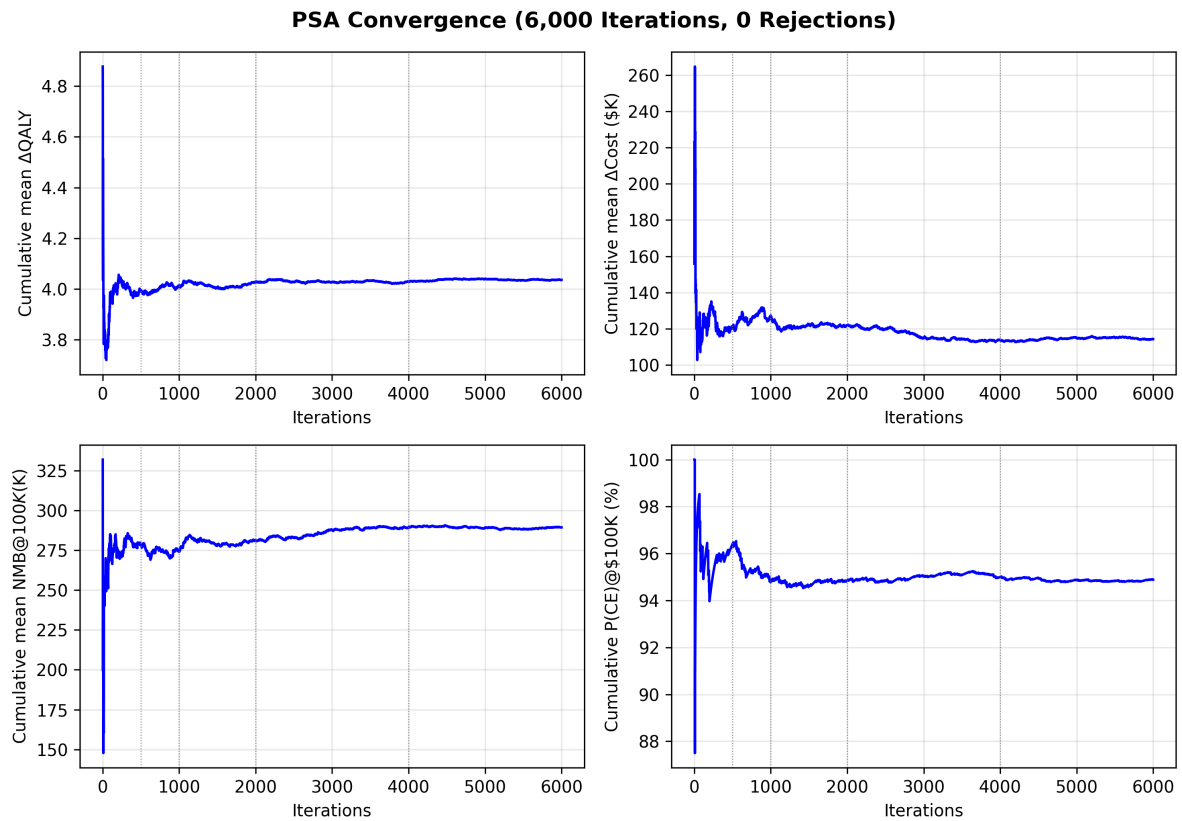


Figure S6: PSA convergence across iteration counts (6,000 structurally valid iterations, 0 structural rejections, seed = 42). Cumulative means of Δ QALY, Δ Cost, NMB, and P(CE). Vertical dotted lines at 500, 1,000, 2,000, and 4,000 iterations. Estimates stable from $\sim$ 500 iterations.

9.2.4 Cost-Effectiveness Plane

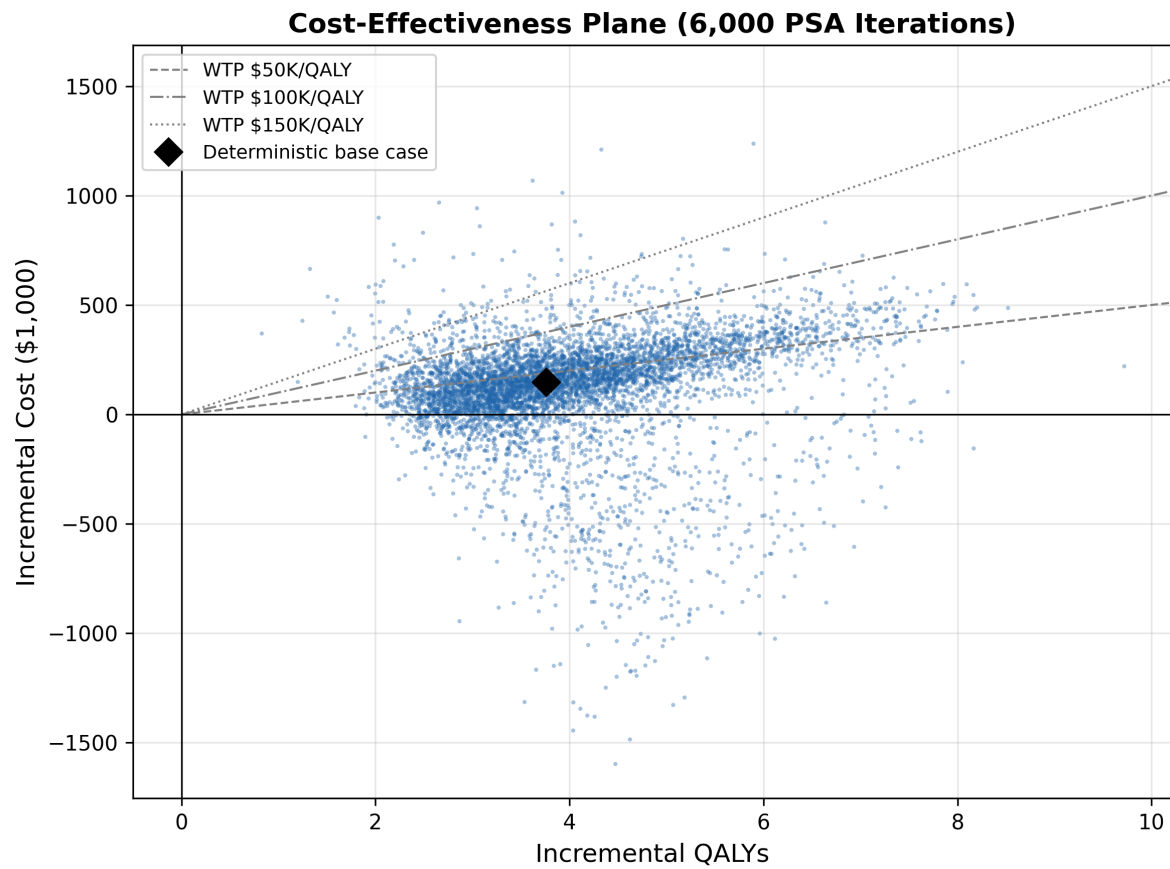


Figure S7: CE plane showing 6,000 PSA iterations (blue), WTP lines (\$50K, \$100K, \$150K per QALY), and the deterministic base case (black diamond). 11.2% of iterations fall in the southeast quadrant (cost-saving); all 6,000 iterations had positive ΔQALY . No iterations were excluded based on the ICER value.

9.3 Structural Scenario Analyses

Table S14: Structural Scenario Analysis Results

Scenario	ΔLY	$\Delta QALY$	$\Delta Cost$	ICER	NMB \$100K	NMB \$150K
Base case	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
Weibull OS	11.05	8.40	\$503,437	\$59,959	\$336,193	\$756,008
Unconstrained (no GP)	9.24	7.22	\$373,488	\$51,753	\$348,189	\$709,028
GP floor 48 mo	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
GP floor 60 mo (base)	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
GP floor 72 mo	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
GP floor 96 mo	8.28	6.53	\$325,801	\$49,862	\$327,608	\$654,313
10-year horizon	1.35	1.34	\$97,868	\$73,233	\$35,771	\$102,591
20-year horizon	4.15	3.52	\$147,215	\$41,785	\$205,099	\$381,256
30-year horizon	6.54	5.29	\$239,745	\$45,349	\$288,922	\$553,256
40-year horizon (base)	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
Intismeran \$100K	8.26	6.52	\$224,765	\$34,494	\$426,840	\$752,643
Intismeran \$200K (base)	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
Intismeran \$300K	8.26	6.52	\$424,765	\$65,187	\$226,840	\$552,643
Intismeran \$500K	8.26	6.52	\$624,765	\$95,881	\$26,840	\$352,643
Waning V1 (recurrence)	2.09	2.09	\$39,862	\$19,055	\$169,330	\$273,926
Waning V2 (hazard convergence)	3.05	2.39	\$249,355	\$104,247	-\$10,159	\$109,439
No direct OS benefit	0.00	0.52	-\$22,543	-\$43,589	\$74,262	\$100,121

The base-case conclusion is robust across structural assumptions. The ICER alone is insufficient: unconstrained model has moderate ICER (\$51,753) but implausible \$373,488 cost.

Time horizon sensitivity. The ICER remained below conventional US willingness-to-pay thresholds across 20-, 30-, and 40-year horizons, although the point estimate increased at the 40-year horizon: \$41,785 at 20 years, \$45,349 at 30 years, and \$49,841 at the 40-year base-case horizon (Table S14). The 30-to-40-year increase of \$4,492/QALY (9.9%) reflects the contribution of late discounted QALY gains that are not fully captured at shorter horizons; the 10-year horizon produced a notably higher ICER of \$73,233/QALY, reflecting the incomplete capture of long-term QALY gains within a shorter window. Because the general-population mortality hazard floor was applied from 60 months onward and both costs and QALYs were discounted at 3% annually, survival beyond the 20-year window contributed materially to discounted incremental costs and QALYs.

General population hazard floor start-time sensitivity. The GP mortality floor was applied from 60 months in the base case. Varying the onset from 48 to 96 months produced no material change in the ICER (all \$49,841/QALY when rounded to whole dollars; the 96-month onset yielded \$49,862/QALY). This reflects the fact that the fitted hazard for the combination arm closely tracks the general-population mortality hazard in the post-trial period (both at approximately 0.0008–0.0009 monthly), leaving the floor with minimal additional constraining effect regardless of onset timing. The floor’s binding is thus a near-numerical coincidence between the parametric extrapolation and US life-table mortality, not a model artifact.

10 Deterministic Threshold Acquisition Price

10.1 Price Threshold Analysis

Table S15: Price Threshold Analysis

Intismeran Price	ICER	NMB @ \$100K	NMB @ \$150K
\$0	\$19,147	\$526,835	\$852,635
\$50,000	\$26,821	\$476,835	\$802,635
\$100,000	\$34,494	\$426,835	\$752,635
\$150,000	\$42,168	\$376,835	\$702,635
\$200,000 (base)	\$49,841	\$326,835	\$652,635
\$250,000	\$57,515	\$276,835	\$602,635
\$300,000	\$65,188	\$226,835	\$552,635
\$350,000	\$72,861	\$176,835	\$502,635
\$400,000	\$80,535	\$126,835	\$452,635
\$450,000	\$88,208	\$76,835	\$402,635
\$500,000	\$95,882	\$26,835	\$352,635
\$526,840	\$100,000	\$0	\$300,000
\$550,000	\$103,555	-\$23,165	\$302,635
\$600,000	\$111,229	-\$73,165	\$252,635
\$650,000	\$118,902	-\$123,165	\$202,635
\$700,000	\$126,575	-\$173,165	\$152,635
\$750,000	\$134,249	-\$223,165	\$102,635
\$800,000	\$141,922	-\$273,165	\$52,635
\$852,643	\$150,000	-\$325,000	\$0

Deterministic threshold prices (from Table S15): approximately \$526,840 at \$100,000/QALY (2.6× base) and approximately \$852,643 at \$150,000/QALY (4.3× base). These are model-based deterministic estimates; PSA uncertainty propagation and formal budget-impact analysis would be needed to inform payer negotiations.

10.2 ICER versus Acquisition Price

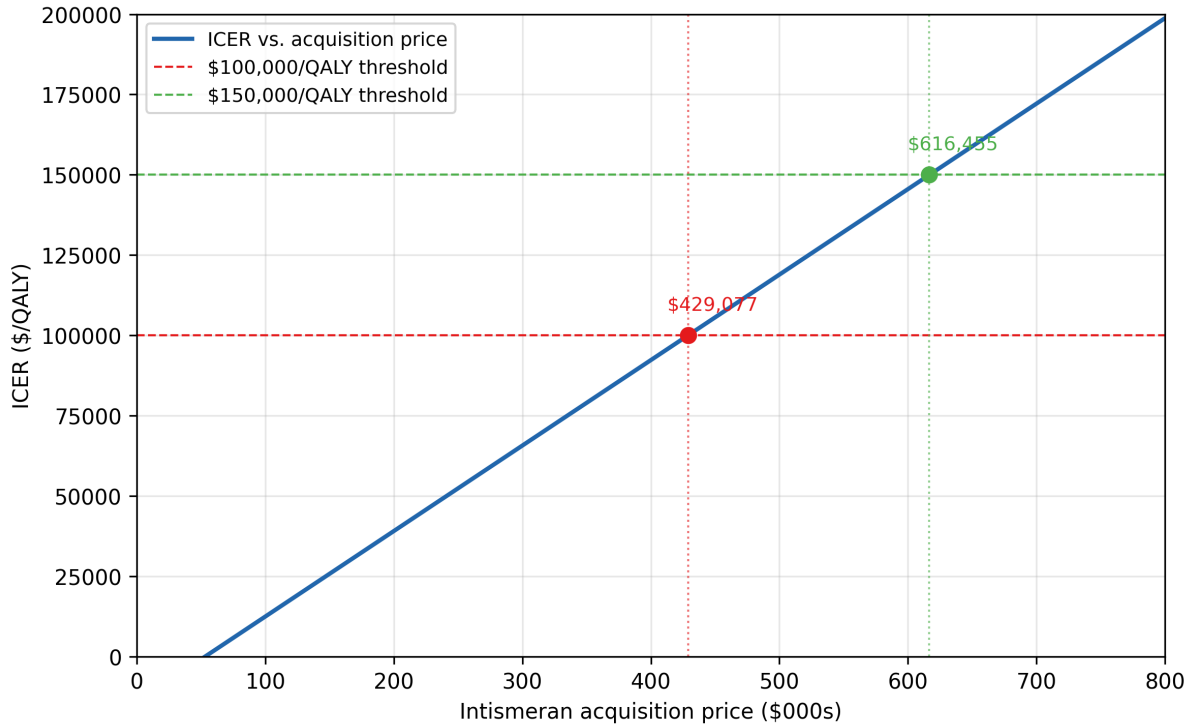


Figure S8: ICER as a function of intismeran acquisition price. Red/green dashed lines: \$100K/\$150K thresholds. Vertical dotted lines: threshold prices.

11 Reproducibility and Reporting

11.1 Technical Reproducibility

All analyses were performed using Python 3.12.7 (Apple Clang 15.0.0) on macOS 26.6.2 (Apple Silicon). The full code repository is available at <https://github.com/yichao2022/cea-intismeran> (tag v1.0-submission).

Key dependencies: NumPy 1.26.4, SciPy 1.14.1, Matplotlib 3.9.3, Pandas 2.2.3, LaTeX (TeX Live 2025, pdfLaTeX + BibTeX + natbib).

Model details:

- Cycle length: monthly (1/12 year); time horizon: 40 years (480 cycles)
- Half-cycle correction: not applied (monthly cycles are sufficiently fine-grained)
- Discounting: continuous-time exponential: $1/(1+r)^t$ where $r = 0.03$
- Random seed: 42 (PSA reproducibility)
- GP mortality source: US 2019 period life tables (NVSS, Arias et al. 2022)

Code structure:

- `model.py` — ModelParams, CEAModel (4-state PSM)
- `rerun_primary.py` — CEAModelV2 (hazard-based GP floor, waning, no-OS)
- `psa_full.py` — PSA with joint survival sampling
- `survival_model_selection.py` — 4,096-combination model selection
- `final_run.py` / `threshold_analysis.py` — DSA, scenarios, pricing
- `figures.py` / `gen_fig*.py` — Figure generation scripts

- `output/survival_combos_4096.csv` — Full model selection results

Data extraction: Survival probabilities extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7. The digitized values are in Table S2.

Reproducibility: Running `python3 rerun_primary.py` reproduces base-case results. `python3 psa_full.py` reproduces PSA. GitHub tag `v1.0-submission` freezes the submission code.

11.2 CHEERS 2022 Checklist

Table S16: CHEERS 2022 Checklist

No.	Item	Reported Location	
1	Title	Yes	Title page
2	Abstract	Yes	Abstract
3	Background	Yes	Introduction
4	Health economic analysis plan	No	This is a retrospective analysis, not a pre-specified HEAP
5	Model	Yes	Methods, Appendix A
6	Population	Yes	Methods
7	Setting	Yes	Methods
8	Comparators	Yes	Methods
9	Perspective	Yes	Methods
10	Time horizon	Yes	Methods
11	Discount rate	Yes	Methods
12	Outcomes	Yes	Methods
13	Health states	Yes	Appendix A
14	Measurement of outcomes	Yes	Methods
15	Resource use	Yes	Methods
16	Currency	Yes	Methods
17	Model choice	Yes	Methods
18	Assumptions	Yes	Methods (§2.2–2.7)
19	Analytic methods	Yes	Methods (§2.7)
20	Study parameters	Yes	Methods (§2.4, 2.5), Appendix D
21	Incremental results	Yes	Results (§3.1–3.2)
22	Uncertainty	Yes	Results (§3.4–3.6), Supplement
23	Heterogeneity	Not assessed	Limitations (§4.7, subgroup heterogeneity acknowledged)
24	Discussion	Yes	Discussion
25	Limitations	Yes	Discussion
26	Funding	Yes	Declarations
27	Conflicts	Yes	Declarations
28	Reproducibility	Yes	Appendix

References

- [1] A. G. Bensimon, Z. Y. Zhou, M. Jenkins, et al. Cost-effectiveness of pembrolizumab for the adjuvant treatment of resected high-risk stage iii melanoma in the united states. *Journal of Medical Economics*, 22(10):981–993, 2019.

- [2] K. M. Johnston et al. Health-state utility values in melanoma. *Pharmacoeconomics*, 39:241–253, 2021.
- [3] S. Zhang, A. G. Bensimon, R. Xu, et al. Cost-effectiveness analysis of pembrolizumab as an adjuvant treatment of resected stage iib or iic melanoma in the united states. *Advances in Therapy*, 40:3038–3055, 2023.
- [4] A. Khattak et al. Five-year follow-up of keynote-942: individualized neoantigen therapy plus pembrolizumab. *Journal of Clinical Oncology*, 44:JCO2600835, 2026.
- [5] J. S. Weber, M. S. Carlino, A. Khattak, et al. Individualised neoantigen therapy mrna-4157 (v940) plus pembrolizumab versus pembrolizumab monotherapy in resected melanoma (keynote-942): a randomised, phase 2b study. *The Lancet*, 403(10421):632–644, 2024.
- [6] P. A. Ascierto, M. Del Vecchio, B. Merelli, et al. Nivolumab for resected stage iii or iv melanoma at 9 years. *New England Journal of Medicine*, 394(4):333–342, 2026. doi: 10.1056/NEJMoA2504966.
- [7] A. M. M. Eggermont, M. Kicinski, C. U. Blank, et al. Seven-year analysis of adjuvant pembrolizumab versus placebo in stage iii melanoma in the eortc1325 / keynote-054 trial. *European Journal of Cancer*, 211:114327, 2024. doi: 10.1016/j.ejca.2024.114327.
- [8] P. A. Ascierto, M. Del Vecchio, M. Mandalá, et al. Adjuvant nivolumab versus ipilimumab in resected stage iii/iv melanoma: 5-year efficacy and biomarker results from checkmate 238. *Clinical Cancer Research*, 29(17):3352–3361, 2023.
- [9] G. D. Sanders, P. J. Neumann, A. Basu, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: Second panel on cost-effectiveness in health and medicine. *JAMA*, 316(10):1093–1103, 2016.
- [10] P. J. Neumann, J. T. Cohen, and M. C. Weinstein. Updating cost-effectiveness — the curious resilience of the \$50,000-per-qaly threshold. *New England Journal of Medicine*, 371(9):796–797, 2014.